Computer Networks Spring 2026
Assignment#6 (6A & 6B) Solution
Part – 1

Problems:

Problem 1

1 1 1 0 1
0 1 1 0 0
1 0 0 1 0
0 1 0 1 0
0 1 0 0 1

Problem 5

If we divide 10011 into 1010101010 0000, we get 1011011100, with a remainder of R=0100. Note that, G=10011 is CRC-4-ITU standard.

Problem 6

a) we get 1001001000, with a remainder of R=1000.

```
              1001001000
10011)10001001010000
      10011
          10001
          10011
              10010
              10011
                  1000
```

b) we get 0101010101, with a remainder of R=1111.

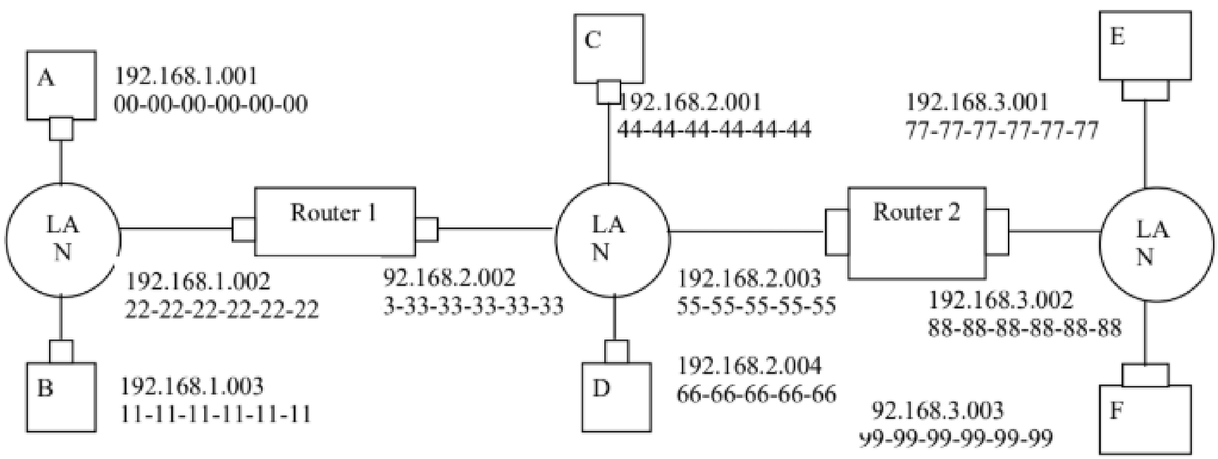
```
             0101010101
10011)01011010100000
       10011
         10110
         10011
           10110
           10011
             10100
             10011
               11100
               10011
                1111
```

c) we get 0110001010, with a remainder of R=1110.

```
           0110001010
10011)01101000110000
         10011
          10010
          10011
              10110
              10011
                10100
                10011
                  1110
```

Problem 14

a), b) See figure below.



c)

1. Forwarding table in E determines that the datagram should be routed to interface 192.168.3.002.
2. The adapter in E creates and Ethernet packet with Ethernet destination address 88-88-88-88-88-88.
3. Router 2 receives the packet and extracts the datagram. The forwarding table in this router indicates that the datagram is to be routed to 198.162.2.002.
4. Router 2 then sends the Ethernet packet with the destination address of 33-33-33-33-33-33 and source address of 55-55-55-55-55-55 via its interface with IP address of 198.162.2.003.
5. The process continues until the packet has reached Host B.

Problem 22

i) from A to switch: Source MAC address: 00-00-00-00-00-00
Destination MAC address: 55-55-55-55-55-55
Source IP: 111.111.111.001
Destination IP: 133.333.333.003

ii) from switch to right router: Source MAC address: 00-00-00-00-00-00
Destination MAC address: 55-55-55-55-55-55
Source IP: 111.111.111.001
Destination IP: 133.333.333.003

iii) from right router to F: Source MAC address: 88-88-88-88-88-88
Destination MAC address: 99-99-99-99-99-99
Source IP: 111.111.111.001
Destination IP: 133.333.333.003

Problem 26

Action	Switch Table State	Link(s) packet is forwarded to	Explanation
B sends a frame to E	Switch learns interface corresponding to MAC address of B	A, C, D, E, and F	Since switch table is empty, so switch does not know the interface corresponding to MAC address of E
E replies with a frame to B	Switch learns interface corresponding to MAC address of E	B	Since switch already knows interface corresponding to MAC address of B
A sends a frame to B	Switch learns the interface corresponding to MAC address of A	B	Since switch already knows the interface corresponding to MAC address of B
B replies with a frame to A	Switch table state remains the same as before	A	Since switch already knows the interface corresponding to MAC address of A

Part – 2

Question 1:

Concept:

If the polynomial is of order n then the number bits generated by CRC generator is n + 1.

Data:

Message = $m_4m_3m_2m_1m_0$ = 11000

CRC polynomial $=X^3+X+1$

Explanation:

CRC polynomial = $1.X^3+ 0.X^2 + 1.X+ 1.X^0 \equiv 1011$

Message bits will be 11000 **000**

Calculation:

```
1011)11000000
    ⊕1011
     01110000
     ⊕1011
       0101000
       ⊕1011
          0100
```

Hence 100 will be appended to message bits ($m_4m_3m_2m_1m_0$ = 11000).

Question 2:

a. CSMA/CD is more efficient because nodes listen before transmitting (avoiding collisions with active transfers) and stop transmitting immediately if a collision is detected (wasting less channel time).
b. Propagation delay. If the delay is high, a node might begin transmitting, not realizing another node far away has already started. This increases the "vulnerable period" for collisions.

Question 3:

Order: B -> E -> D -> A -> C -> F

1. **B & E (DHCP):** The laptop needs an IP address and DNS server info before doing anything else.
2. **D (ARP for Gateway):** Before sending the DNS query (which goes to a server typically outside the LAN), the laptop needs the MAC address of the Gateway to send the packet out.
3. **A (DNS):** The laptop needs the IP of google.com to address the TCP connection.
4. **C (TCP Handshake):** Once IP is known, a connection is established (SYN, SYN-ACK, ACK).

5. **F (HTTP):** Only after the connection is established can the HTTP GET request be sent.

Question 4:

a. For Node A to succeed in a specific slot, two conditions must be met simultaneously:
   1. Node A must transmit (probability p).
   2. All other nodes (B, C, and D) must **not** transmit (probability $1-p$ for each).

   Calculation:
   $P(\text{A succeeds})=p\times(1-p)^{(N-1)}$
   $P(\text{A succeeds})=0.2\times(1-0.2)^3$
   Answer: 0.1024 (or 10.24%)

b. For Node A to succeed for the *first time* in slot 4, it must fail to succeed in slots 1, 2, and 3, and then succeed in slot 4.

   Step 1: Determine the probability of A failing to succeed in a single slot. We already know the probability of success is 0.1024. Therefore, the probability of *not* succeeding is:

   $P(\text{A fails})=1-P(\text{A succeeds})$
   $P(\text{A fails})=1-0.1024=0.8976$

   Step 2: Calculate the sequence of events.
   - Slot 1: Fail (0.8976)
   - Slot 2: Fail (0.8976)
   - Slot 3: Fail (0.8976)
   - Slot 4: Succeed (0.1024)

   Since the slots are independent, we multiply the probabilities:
   $P(\text{Success in Slot 4})=P(\text{fail})^3\times P(\text{succeed})$
   $P=(0.8976)^3\times0.1024$
   Answer: Approximately **0.074** (or 7.4%)

Question 5:

Packet 1		
Header	**Source**	**Destination**
Ethernet	MAC 16	MAC 15
IP Address	3.0.1.2	1.2.3.4

Packet 2		
Header	**Source**	**Destination**
Ethernet	MAC 1	MAC 2
IP Address	10.0.0.2	3.0.1.2

Packet 3		
Header	**Source**	**Destination**
Ethernet	MAC 3	MAC 4
IP Address	1.2.3.4	3.0.1.2

Packet 4		
Header	**Source**	**Destination**
Ethernet	MAC 9	MAC 10
IP Address	1.2.3.8	1.2.4.3
IP Address	1.2.3.6	2.0.0.1
IP Address	1.2.3.4	3.0.1.2